

PopBeauty Inventory Analytics

Diagnosing and solving stock misallocation across a UK retail network

About this dashboard: This is a portfolio project built on a *synthetic dataset* created for an academic group assignment. All figures are illustrative and demonstrate analytical technique — they do not represent real company performance. Because the data is synthetic, this dashboard can be shared freely, unlike production BI work under NDA.

Purpose: This dashboard analyses a fictional UK cosmetics retailer's inventory across stores and warehouses, identifying where stock is misallocated, at risk of expiry, or stuck in the supply chain — and recommends specific, costless corrective actions. It was built to demonstrate end-to-end BI development: from a relational database to an interactive, decision-ready dashboard.

Technical skills demonstrated:

- SQLite database connection via ODBC
- Star-schema data modelling (12 relationships, multiple fact tables)
- DAX measures (time intelligence, ratios, calculated columns)
- Multi-page interactive design with cross-filtering and drill-down
- Translating data into actionable business recommendations

The five pages:

1. **Network Overview** — where stock is misallocated across the network
2. **Premium Misallocation** — high-value stock trapped in low-traffic stores
3. **Expiry Risk & Redistribution** — surplus stock approaching expiry
4. **Fulfilment & Reorder** — root causes: stuck shipments & mis-set reorder levels
5. **Warehouse vs Store Shortage** — stock pooled upstream while shelves run thin

How to navigate: Use the left panel to move between pages. Each page diagnoses one dimension of the inventory problem. Use the slicers to filter, and click any visual to cross-filter the others.

Context: Built [Month Year] as part of an MSc Business Analytics project at Warwick Business School. The underlying database was designed by my project team; this Power BI dashboard is my individual work, extending that database into an interactive BI tool.

Network Overview: Inventory Misallocation Diagnosis

Start Here

Network Overview

Premium Misallocation

Expiry Risk & Redistribution

Fulfilment & Reorder (Root Cause)

Warehouse Stock vs Store Shortage

Filters

region

All

store_size_category

All

category_name

All

£568K

Total Revenue

£284.89K

Gross Profit

50.17%

Gross Margin %

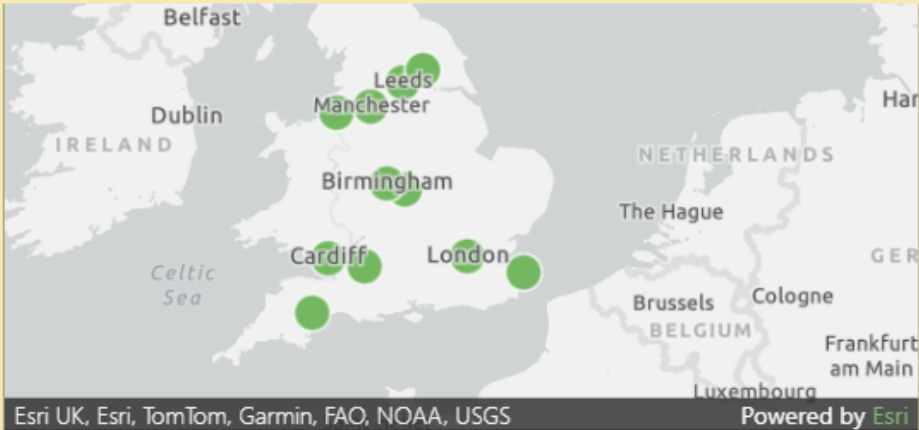
162

High-Risk Expiry Count

135

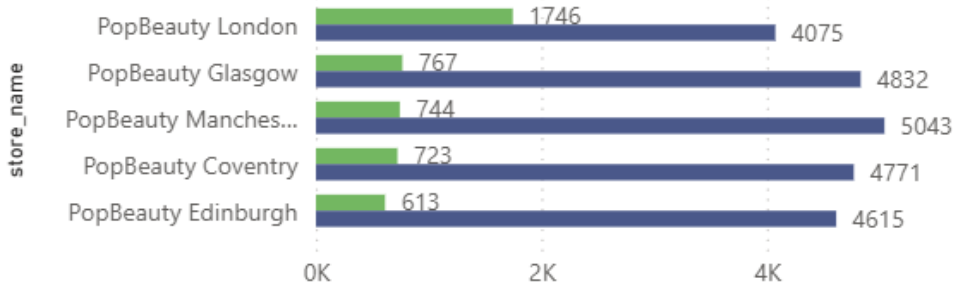
Stockout Situations

Sell-Through by Store (Geographic)

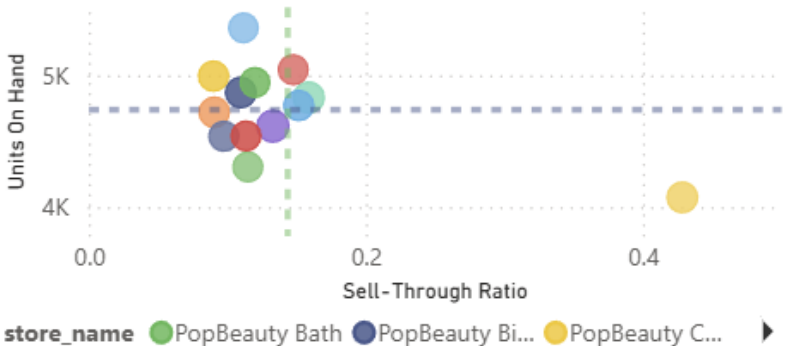


Units Sold and Units On Hand by stores

● Units Sold ● Units On Hand



Store Positioning — Sell-Through vs Margin



store_name	Units On Hand	Units Sold	Sell-Through Ratio	Gross Margin %
PopBeauty York	5360	598	0.11	47.02%
PopBeauty Manchester	5043	744	0.15	48.24%
PopBeauty London	4075	1746	0.43	51.05%
PopBeauty Liverpool	4538	442	0.10	48.43%
PopBeauty Leeds	4305	494	0.11	48.11%
PopBeauty Glasgow	4832	767	0.16	53.00%
PopBeauty Exeter	4720	427	0.09	50.12%
Total	61609	8645	0.14	50.17%

Stock isn't short — it's misallocated. London converts 43% of stock; boutiques 9%, while holding the most units.

Premium Misallocation : High Volume Stock trapped in Low Traffic Stores

Start Here

Network Overview

Premium Misallocation

Expiry Risk & Redistribution

Fulfilment & Reorder (Root Cause)

Warehouse Stock vs Store Shortage

Filters

region

All

store_size_category

All

£291.15K

Trapped Premium Stock Value

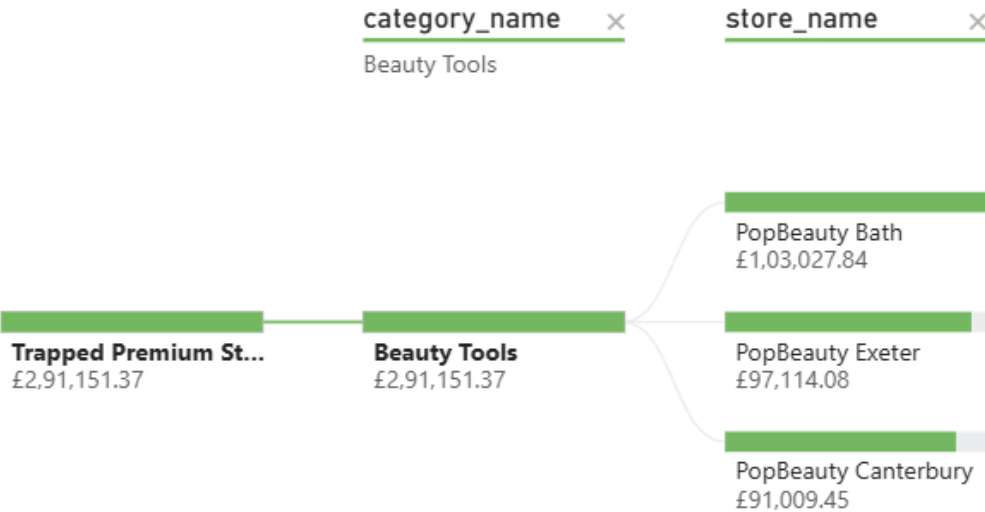
1,373

Premium Units in Boutiques

7.43%

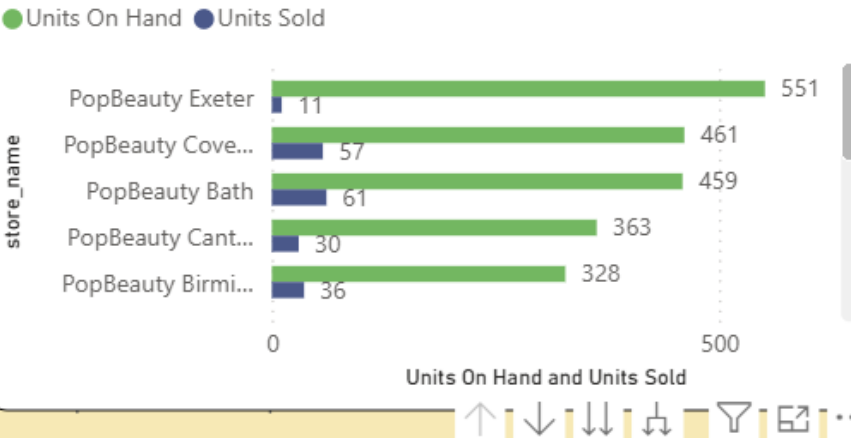
Boutique Premium Sell-Through

Where is Trapped Premium Value Concentrated?



Key Insight: £291K of premium Beauty Tools (at retail value) sits unsold in boutiques — Exeter holds 551 units and sold just 11, while London sells 116 of 160. Because Beauty Tools earn the network's lowest category margin (39%), this idle stock ties up significant capital for thin returns, making redistribution urgent. **Action: move premium stock from boutiques to the Flagship and high-velocity stores.**

Premium Beauty Tools — Stocked vs Sold.



store_name	Units On Hand	Units Sold	Beauty Tools
PopBeauty Bath	459	61	
PopBeauty Birmingham	328	36	
PopBeauty Canterbury	363	30	
PopBeauty Cardiff	313	39	
PopBeauty Coventry	461	57	
PopBeauty Edinburgh	66	34	
PopBeauty Exeter	551	11	
PopBeauty Glasgow	207	50	
PopBeauty Leeds	294	18	
PopBeauty Liverpool	198	7	
Total	3762	541	

Expiry Risk & Redistribution: Where at-risk stock Concentrates

Start Here

Network Overview

Premium Misallocation

Expiry Risk & Redistribution

Fulfilment & Reorder (Root Cause)

Warehouse Stock vs Store Shortage

Filters

region

All

store_size_category

All

category_name

All

£764K

Estimated Write-off Value

162

High-Risk Expiry Count

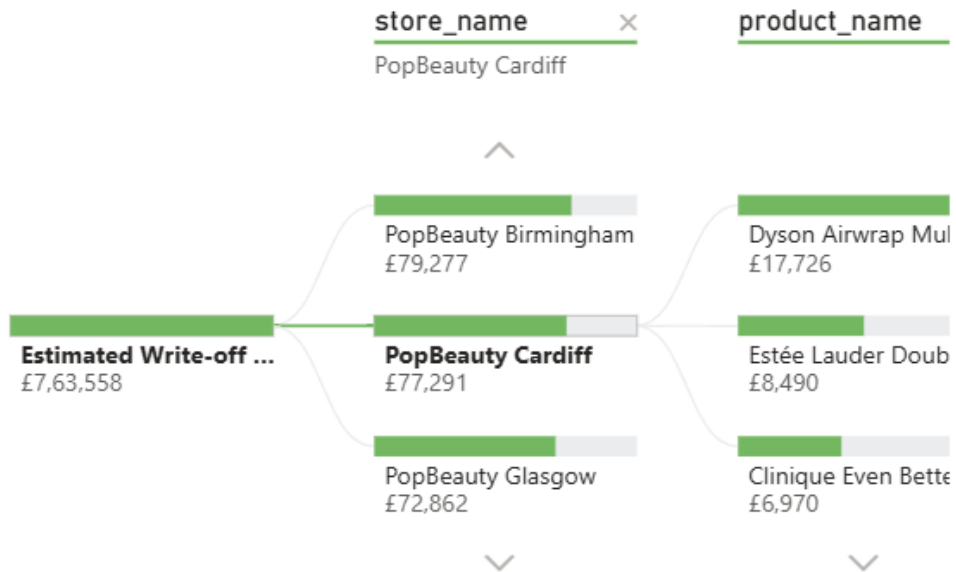
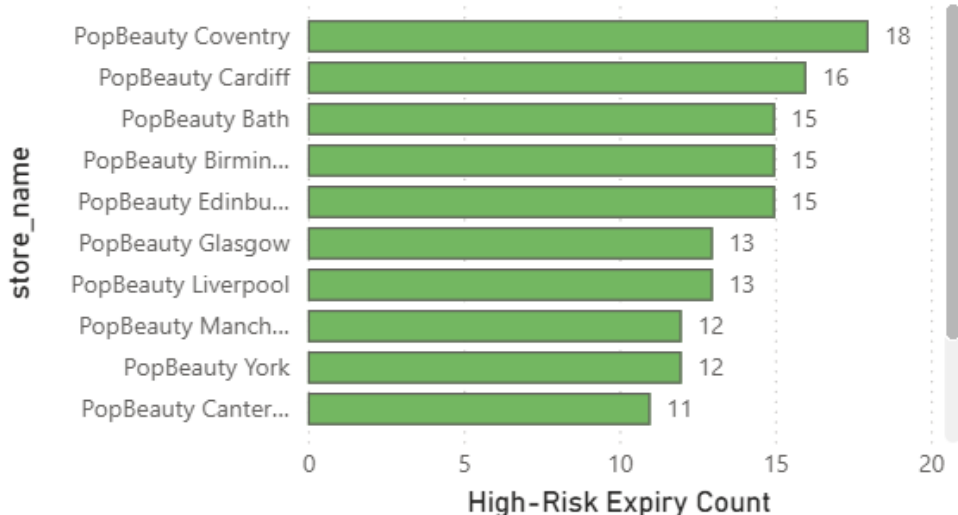
232

Critical Shortage Situations

22.35%

Critical Shortage %

High-Risk Expiry Units by Store.



Same SKUs: Surplus & Expiring vs Critically Short by Region.

product_name	North West	Scotland	South East	South West	Wales	West Midlands	Yorkshire	Total
Benefit Benetint - Hope				21				21
Benefit Boi-ing Cakeless Concealer - Light 1					29	26		55
Benefit Boi-ing Cakeless Concealer - Tan 1		81				85		166
Benefit Hoola Matte Bronzer - Medium						15		15
Benefit The POREfessional Face Primer - 30ml		5						5
Bobbi Brown Vitamin Enriched Face Base - Mini 15ml			68		0			68
Briogeo Don't Despair, Repair! Deep Conditioning Mask - 200ml	58					48		106
Briogeo Don't Despair, Repair! Deep Conditioning Mask - 250ml		72						72
Total	1313	1381	850	1100	616	1377	960	7597

Fulfilment & Reorder: Ghost Shipments and Mis-Set Reorder Levels

Start Here

Network Overview

Premium Misallocation

Expiry Risk & Redistribution

**Fulfilment & Reorder
(Root Cause)**

Warehouse Stock vs Store Shortage

Filters

region

All

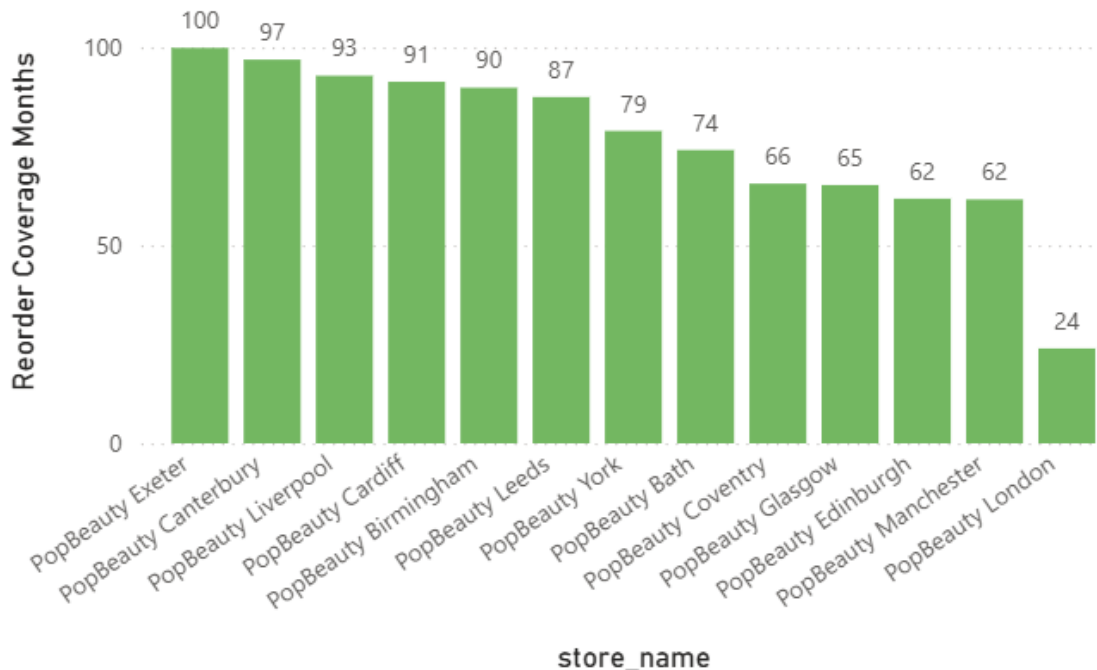
store_size_category

All

category_name

All

Reorder Coverage (Months) by Store.

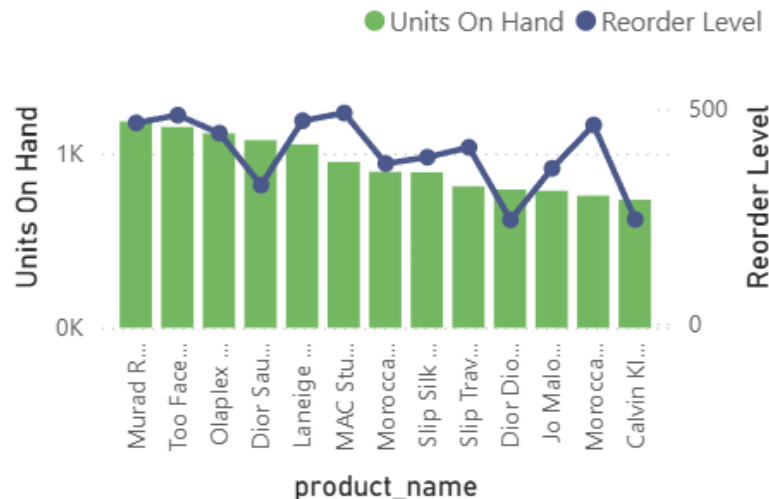


Stock vs Reorder Level (select a store).

store_name

All

Stock vs Reorder Level — Selected Store



Ghost Shipments — Stuck in Transit

shipment_id	store_name	product_name	Sum of quantity	Sum of Days In Transit	shipment_date
SHP00220	PopBeauty Manchester	CHANEL Coco Mademoiselle Eau de Parfum - 50ml	21	442	08-01-2025
SHP00423	PopBeauty Exeter	Clinique All About Eyes - 20ml	92	441	09-01-2025
SHP00878	PopBeauty Leeds	Moroccanoil Color Depositing Mask - 150ml	97	439	11-01-2025
SHP00166	PopBeauty London	FOREO LUNA 4 - Mint	88	438	12-01-2025
SHP00989	PopBeauty Liverpool	Rare Beauty Liquid Touch Brightening Concealer - Medium 1	105	428	22-01-2025
SHP00583	PopBeauty Glasgow	Too Faced Born This Way Foundation - 330N	44	422	28-01-2025
SHP00429	PopBeauty Glasgow	Olaplex No.4 Bond Maintenance Shampoo - 500ml	101	419	31-01-2025
SHP00729	PopBeauty Birmingham	MAC Studio Fix Fluid SPF 15 Foundation - 260N	30	416	03-02-2025
Total			64792	39454	

Warehouse Stock vs Store Shortage: Full Warehouses, Empty Shelves

Start Here

Network Overview

Premium Misallocation

Expiry Risk & Redistribution

Fulfilment & Reorder (Root Cause)

Warehouse Stock vs Store Shortage

Filters

region

All

store_size_category

All

category_name

All

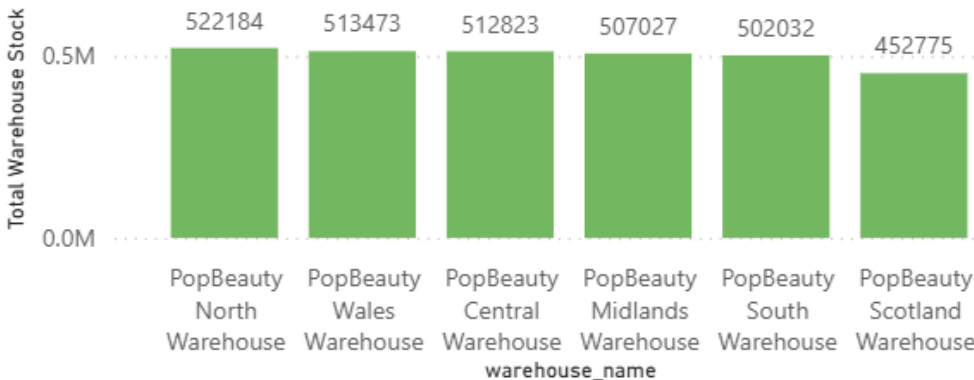
3010314

Total Warehouse Stock

134

Stuck Upstream Count

Stock Held by Warehouse.



Key Insight: Over 3 million units sit in warehouses — evenly spread across all six sites (~0.5M each) — while stores run thin. 134 products are well-stocked upstream yet below reorder level in stores: Beautyblender holds 34,651 units in warehouse but just 459 on shelves. The constraint is distribution, not supply — the stock exists, it isn't reaching the shelf. **Action: release warehouse stock to shortage stores before placing new reorders.**

Warehouse Stock vs Store Shortage (Same Products)

product_name

Total Warehouse Stock Average

Beautyblender Original Beautyblender - Original

34651

Benefit Benetint - Hope

26980

Benefit Boi-ing Cakeless Concealer - Light 1

23528

Benefit Boi-ing Cakeless Concealer - Tan 1

28740

Benefit Hoola Matte Bronzer - Light

14589

Benefit Hoola Matte Bronzer - Medium

15590

Benefit The POREfessional Face Primer - 30ml

14236

Benefit The POREfessional Face Primer - 40ml

19699

Bobbi Brown Vitamin Enriched Face Base - 30ml

17068

Bobbi Brown Vitamin Enriched Face Base - Mini 15ml

13442

Total

3010314